Battery-Efficient & Representation-Fair Federated Learning Research Blueprint (v2.0)

An Edge-Resource Adaptive Framework with FedNova, INT8 Quantization & Participation Fairness
Target Publication Track: Option A (IEEE Access / Workshop / Regional Conference) | 3-Month Plan

1. Executive Summary & Validated Research Scope

Federated Learning (FL) enables distributed edge devices (smartphones, IoT sensors) to collaboratively train global machine learning models without transferring raw data. However, real-world edge devices exhibit severe resource heterogeneity: varied battery levels (15% to 100%), differing network connectivity, and unpredictable dropouts. Naive greedy battery-aware client selection (e.g., EAFL) creates severe demographic representation bias under non-IID data distributions, silencing budget devices.

Furthermore, allowing devices to train with varying local epochs without aggregation correction introduces Objective Inconsistency (Client Drift), causing model divergence (Wang et al., NeurIPS 2020).

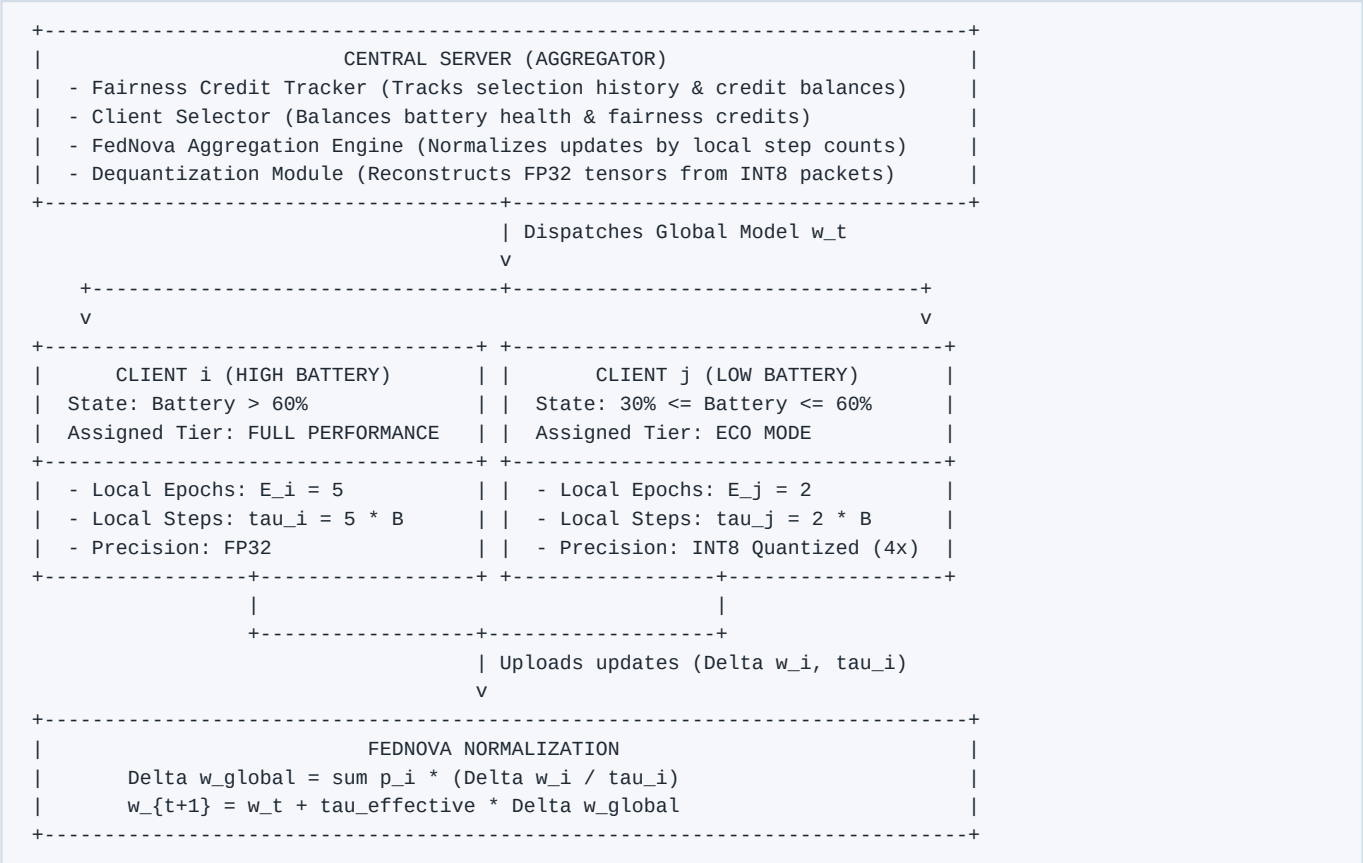
Our upgraded framework reconciles energy efficiency, mathematical convergence, and fairness by coordinating:

* **Rule-Based Battery Governor:** Dynamically places devices into Full, Eco, or Sleep tiers per round based on live battery.
* **FedNova Aggregation Engine:** Normalizes client updates by local iteration counts, mathematically preventing client drift.
* **INT8 Uniform Quantization:** Compresses parameter uploads by 75% (4x data reduction) without high edge CPU sort overhead.
* **Participation Fairness Credit:** Tracks client selection counts to ensure chronically low-battery clients are not starved.

2. Core Research Questions (RQs)

* **RQ1 (Energy vs. Accuracy):** Can adaptive local epochs combined with INT8 quantization reduce total energy consumption by >= 40% with <= 1.5% Top-1 accuracy loss vs. vanilla FedAvg?
* **RQ2 (Mathematical Convergence):** Does FedNova eliminate the objective inconsistency and client drift observed when standard FedAvg is paired with variable local epochs?
* **RQ3 (Communication Efficiency):** What are the empirical bandwidth savings (MB transferred) of INT8 quantization across communication rounds?
* **RQ4 (Fairness & Bias Audit):** Does greedy battery selection cause systematic class accuracy drops on Dirichlet non-IID splits, and does our Fairness Credit restore fair representation (Jain's Fairness Index >= 0.85)?

3. System Architecture & Operational Pipeline



Operational Tier Rules:

* **Full Performance Tier (Battery > 60%):** Full local training (E = 5 epochs), standard FP32 precision.
* **Eco Tier (30% <= Battery <= 60%):** Reduced training (E = 2 epochs), INT8 quantization applied to model updates.
* **Sleep Tier (Battery < 30%):** Preserves device power; participates only when Fairness Credit exceeds threshold.

4. Mathematical Foundations

4.1 FedNova Normalized Aggregation

Standard FedAvg computes Delta w = sum p_i * Delta w_i. When clients execute different local epochs (E_i), local progress tau_i varies, pulling the global model toward high-epoch clients. FedNova normalizes updates by local progress: d_i = Delta w_i / tau_i. The global update is weighted by effective steps: Delta w_global = tau_eff * sum(p_i * d_i), preserving objective consistency and ensuring true convergence.

4.2 INT8 Uniform Symmetric Quantization

Model update tensors are mapped into signed 8-bit integers: Scale S = max(|Delta w|) / 127. Quantized weights q = clamp(round(Delta w / S), -127, 127). The client transmits q and scalar S, reducing transmission bandwidth by 75% in O(M) linear time with minimal edge compute.

4.3 Energy Consumption Model

Total round energy per client: E_total = E_comp + E_comm. Compute energy E_comp = E_i * |D_i| * C_model * kappa * f^2. Communication energy E_comm = P_tx * (Bytes_sent / Data_rate). Battery discharge is decremented accordingly each round.

4.4 Jain's Fairness Index

To audit selection equity across all K clients, we evaluate: J(x) = (sum x_i)^2 / (K * sum x_i^2), where x_i is the cumulative participation count of client i. J in [1/K, 1.0], with 1.0 representing ideal uniform participation.

5. Team Roles & Responsibilities (6 Members)

To ensure balanced workload, equal contribution, and clear authorship sections for the IEEE paper:

Member	Assigned Title	Core Deliverables & Code	Paper Section
Member 1	Tech Lead & System Architect	Flower simulation runner, client-server loop, YAML config	Sec III: System Architecture
Member 2	Aggregation & Optimization Lead	Custom FedNova Flower Strategy, local step normalizatic	Sec IV: FedNova Convergence
Member 3	Compression & Quantization Lead	INT8 symmetric quantization/dequantization, bandwidth n	Sec V: Quantization & Comm.
Member 4	Battery & Energy Modeling Lead	Energy equations (E_comp + E_comm), battery state sim	Sec VI: Energy Modeling
Member 5	Adaptive Controller & Fairness Lead	3-tier rule governor, Fairness Credit accumulator, Jain's F	Sec VII: Adaptive Governor
Member 6	Benchmarking & Paper Lead	CIFAR-10 Dirichlet non-IID loader, 4 baseline runs, IEEE	Abstract, Intro, Sec VIII: Results

6. 3-Month (12-Week) Execution Roadmap

Month 1: Foundations & Independent Prototyping (Weeks 1 - 4)

* **Week 1 (Setup):** Setup Python 3.11, PyTorch, Flower (flwr). M1 sets up repo skeleton; M6 builds CIFAR-10 CNN.
* **Week 2 (Data & Energy):** M6 creates Dirichlet non-IID split (alpha=0.5). M4 builds battery.py simulator. M1 runs vanilla FedAvg.
* **Week 3 (Algorithms):** M3 codes quantization.py. M2 prototypes FedNova strategy in Flower. M5 codes rule-based governor.
* **Week 4 (Unit Verification):** M5 adds Fairness Credits. M3 validates INT8 accuracy impact (<0.5%). Month 1 Milestone met.

Month 2: Integration & Baseline Execution (Weeks 5 - 8)

* **Week 5 (Core Hookup):** Hook controller and FedNova into Flower fit_round lifecycle (M1, M2, M5). Add INT8 to client.py (M3).
* **Week 6 (Baselines 1 & 2):** Run Baseline 1 (Vanilla FedAvg) & Baseline 2 (Greedy EAFL Selection) for 50 rounds. Log accuracy & energy.
* **Week 7 (Baselines 3 & Proposed):** Run Baseline 3 (Heterogeneous FedAvg drift test). Run Proposed PAC-FL (FedNova + INT8 + Fairness).
* **Week 8 (Data Consolidation):** Consolidate CSV metrics into Pandas dataframes. Validate accuracy within 1.5% and energy savings >= 40%.

Month 3: Ablations, Visualization & Paper Writing (Weeks 9 - 12)

* **Week 9 (Ablation Runs):** Run ablation experiments: Proposed without INT8; Proposed without Fairness Credits under heavy non-IID (alpha=0.1).
* **Week 10 (Publication Plots):** Generate Pareto Frontier (Accuracy vs. Joules), Participation histograms, and Bandwidth MB charts.
* **Week 11 (Paper Drafting):** Compile complete IEEE format paper in LaTeX. Each of the 6 members writes designated sections.
* **Week 12 (Final Polish):** Peer review, reference verification, presentation slides, code release on GitHub. Ready for submission.

7. Experimental Design & Benchmarking Protocol

All experiments are performed on CIFAR-10 across 20-50 simulated edge clients partitioned via Dirichlet non-IID distributions:

* **Baseline 1 (Vanilla FedAvg):** Uniform client selection, fixed E=5, standard FP32 precision. Represents accuracy and energy upper bound.

* **Baseline 2 (Greedy EAFL):** Selects only top-battery clients (>=50%), fixed E=5. Demonstrates demographic starvation on non-IID data.

* **Baseline 3 (Unnormalized FedAvg):** Variable local epochs (E_i) based on battery, aggregated via standard FedAvg. Proves client drift failure.

* **Proposed Framework (PAC-FL):** Adaptive local epochs (E_i) + INT8 quantization + FedNova aggregation + Participation Fairness Credits.

Target Success Metrics:

* **Top-1 Accuracy:** Final accuracy within <= 1.5% of Vanilla FedAvg upper bound.

* **Energy Reduction:** At least 40% reduction in aggregate Joules consumed across clients.

* **Bandwidth Reduction:** At least 70% reduction in total megabytes uploaded across rounds.

* **Jain's Fairness Index:** J >= 0.85 (compared to J <= 0.45 under greedy battery selection).

8. Key References for IEEE Publication

1. Beutel, D. J. et al. (2020). Flower: A Friendly Federated Learning Framework. arXiv:2007.14390.
2. Wang, J. et al. (2020). Tackling Objective Inconsistency in Heterogeneous Federated Optimization (FedNova). NeurIPS 33.
3. Arouj, A. et al. (2022). Towards Energy-Aware Federated Learning on Battery-Powered Clients (EAFL). ACM FedEdge.
4. Savoia, M. et al. (2024). Eco-FL: Enhancing Federated Learning Sustainability. Computer Communications, 225.
5. Dettmers, T. et al. (2022). 8-bit Optimizers via Block-wise Dynamic Quantization. ICLR.
6. Marnissi, O. et al. (2025). FairEnergy: Contribution-Based Fairness meets Energy Efficiency in FL. arXiv:2511.15454.
7. Li, T. et al. (2020). Federated Optimization in Heterogeneous Networks (FedProx). MLSys.
8. Jain, R. et al. (1984). A Quantitative Measure of Fairness for Resource Allocation. DEC Tech Report.